

Temporal Pattern Detection

Week-2

Task-3 (DS-3)

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1. Dataset:

- Office for National Statistics (ONS) - Crime in England and Wales: Annual Supplementary Tables, March 2025 edition is used.
- The file contains 54 sheets (Tables A, B, C series). Two tables were identified as mental health proxies and extracted:
 - Table B4 - % of people with high levels of worry about crime (anxiety proxy) | 1992–2025.
 - Table B7 - % of people feeling safe walking alone after dark (fear proxy) | 1994–2025.
- A custom time series CSV was constructed by merging both tables on year, applying linear interpolation for B7 2017 (single missing point), and flagging 2021–2022 as COVID gap (no survey conducted).
- Fraud worry (Table B4) excluded — only 6 data points (2019–2025), insufficient for time series analysis.

(29, 7)		year	burglary_worry_pct	car_crime_worry_pct	violent_crime_worry_pct	feel_safe_male_pct	feel_safe_female_pct	feel_safe_all_pct
0	1992	19.0000	NaN	NaN	NaN	NaN	NaN	NaN
1	1994	26.0000	NaN	NaN	84.0978	45.4771	64.0487	
2	1996	22.1255	NaN	NaN	85.0083	52.6589	68.2005	
3	1998	19.3000	22.3	24.6	85.0804	51.5994	67.7154	
4	2000	19.2000	21.0	23.7	84.9843	52.7326	68.2818	

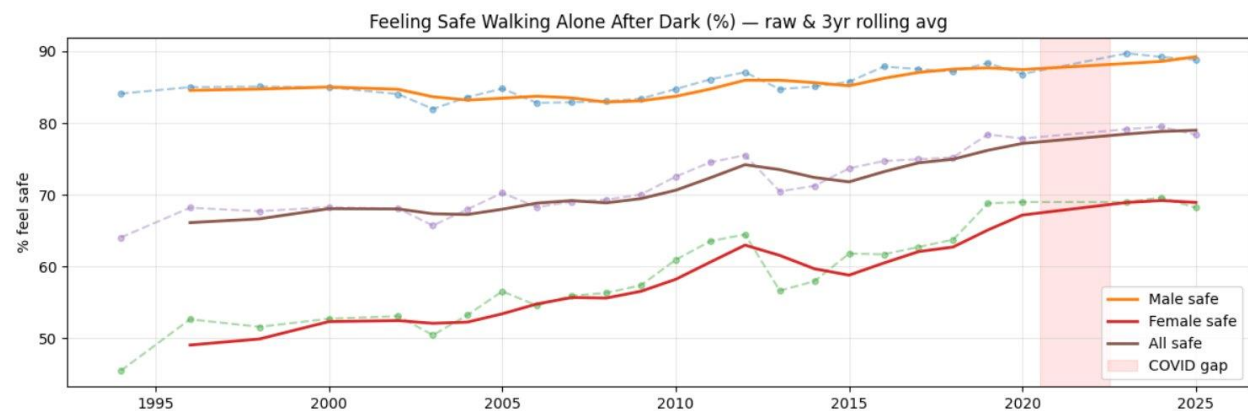
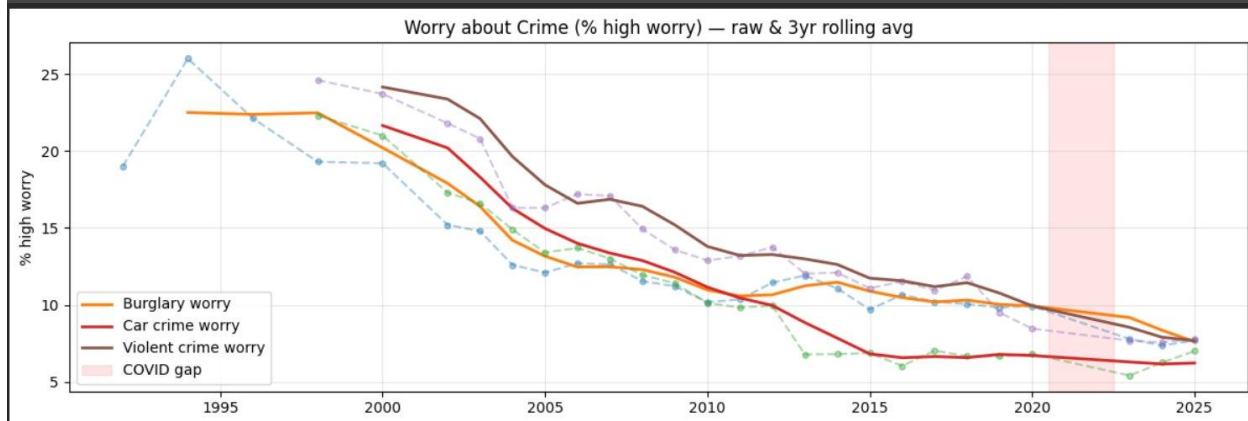
2. Data Overview:

- Final dataset: 29 rows, 7 columns (including 2 COVID gap rows).
- Annual series used for analysis (2002 onwards): 22 rows.
- Variables:
 - burglary_worry_pct, car_crime_worry_pct, violent_crime_worry_pct (Table B4).
 - feel_safe_male_pct, feel_safe_female_pct, feel_safe_all_pct (Table B7).
- Pre-2002 data is biennial (1992, 1994, 1996, 1998, 2000) — used in rolling averages, excluded from Chow test.

3. Rolling Averages (3-year window):

- Applied 3-year rolling mean to all 6 variables (min_periods=2 to handle biennial start).
- COVID rows (2021, 2022) dropped before rolling calculation to avoid artificial flatlines.
- Rolling averages smooth short-term volatility and reveal structural trends clearly.

	year	burglary_worry_pct_ra3	car_crime_worry_pct_ra3	violent_crime_worry_pct_ra3	feel_safe_male_pct_ra3	feel_safe_female_pct_ra3	feel_safe_all_pct_ra3
0	1992	NaN	NaN	NaN	NaN	NaN	NaN
1	1994	22.5000	NaN	NaN	NaN	NaN	NaN
2	1996	22.3752	NaN	NaN	84.5531	49.0680	66.1246
3	1998	22.4752	NaN	NaN	84.7288	49.9118	66.6549
4	2000	20.2085	21.6500	24.1500	85.0243	52.3303	68.0659
5	2002	17.9000	20.2000	23.3667	84.7001	52.4762	68.0278
6	2003	16.4000	18.3000	22.1000	83.6621	52.0923	67.3610
7	2004	14.2000	16.2667	19.6333	83.2019	52.2675	67.2615
8	2005	13.1667	14.9667	17.8000	83.4577	53.4189	67.9869
9	2006	12.4667	14.0000	16.6000	83.7407	54.7926	68.8415
10	2007	12.4798	13.3702	16.8603	83.5029	55.6894	69.1934
11	2008	12.2953	12.8835	16.4032	82.9145	55.6173	68.8799
12	2009	11.7993	12.1193	15.1957	83.1070	56.5584	69.4691
13	2010	10.9787	11.1566	13.7985	83.7231	58.2235	70.6340
14	2011	10.5830	10.4560	13.2143	84.7392	60.6267	72.3724
15	2012	10.6613	9.9682	13.2732	85.9690	62.9900	74.1954



4. Structural Break Detection (Chow Test):

- Method: Chow test comparing OLS regression fit in two sub-periods split at a break year.
- Formula: $F = [(RSS_{\text{pooled}} - RSS1 - RSS2) / k] / [(RSS1 + RSS2) / (n - 2k)]$, where $k=2$.
- Annual series only (2002–2025) used for consistent annual spacing required.
- Two approaches run: fixed break at 2013, and full scan across all possible break years.

4.1 Fixed Break Test (2013):

variable	break_year	F_stat	p_value	significant
burglary_worry_pct	2013	1.2059	0.3225	NO
car_crime_worry_pct	2013	57.1520	0.0000	YES
violent_crime_worry_pct	2013	3.9823	0.0370	YES
feel_safe_male_pct	2013	0.1694	0.8455	NO
feel_safe_female_pct	2013	0.8667	0.4372	NO
feel_safe_all_pct	2013	0.4081	0.6709	NO

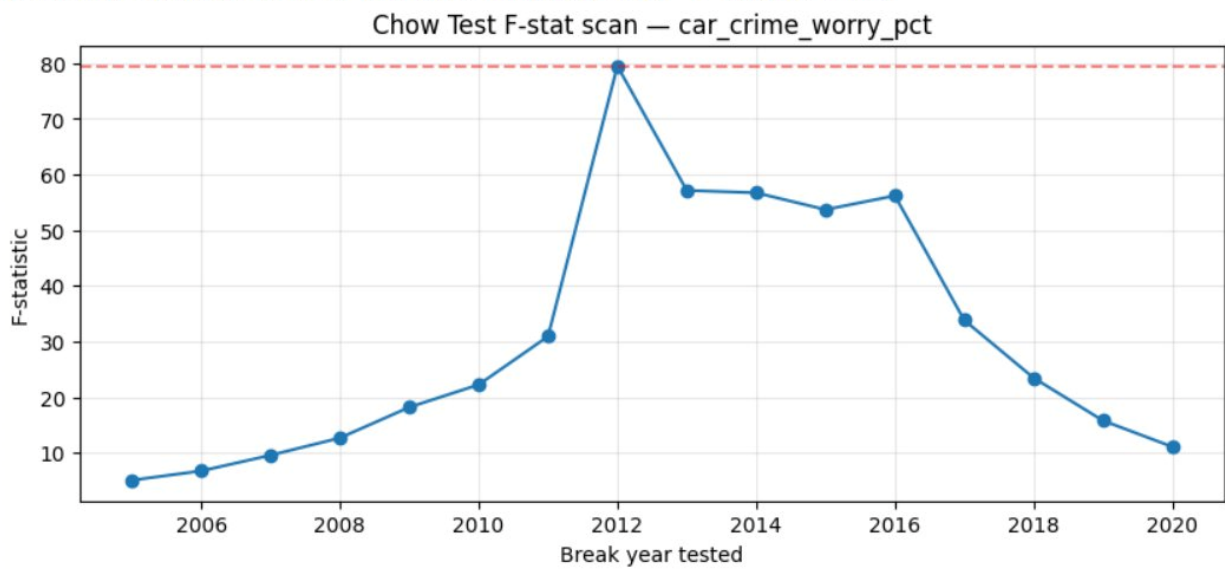
- Car crime worry: $F=57.15$, $p=0.000$ - **SIGNIFICANT**.
- Violent crime worry: $F=3.98$, $p=0.037$ - **SIGNIFICANT**.
- Burglary worry, all safety variables: **not significant at 2013**.

4.2 Break Year Scan (Most Likely Structural Break):

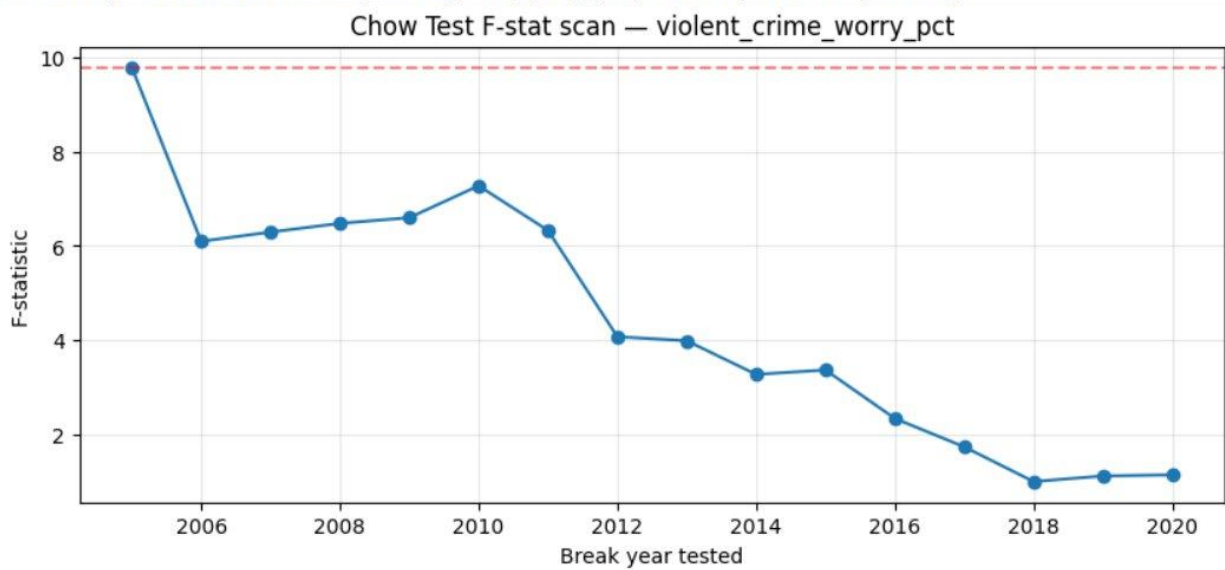
- Scanned all possible break years; identified year with highest F-statistic per variable



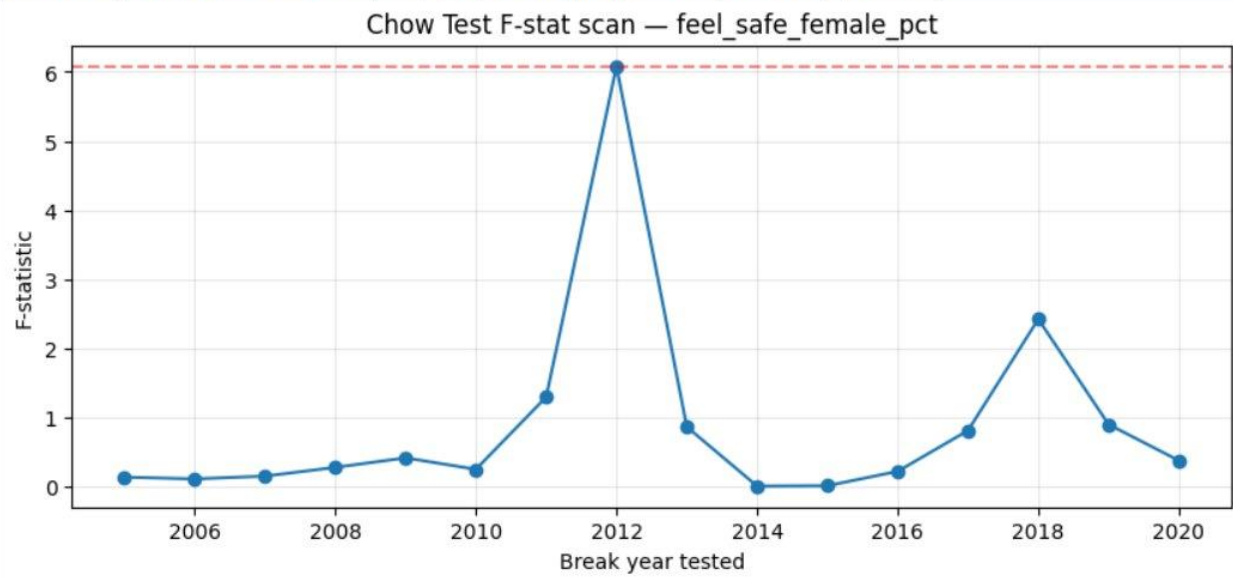
Most likely structural break in [car_crime_worry_pct]: 2012 ($F=79.4818$, $p=0.0$)



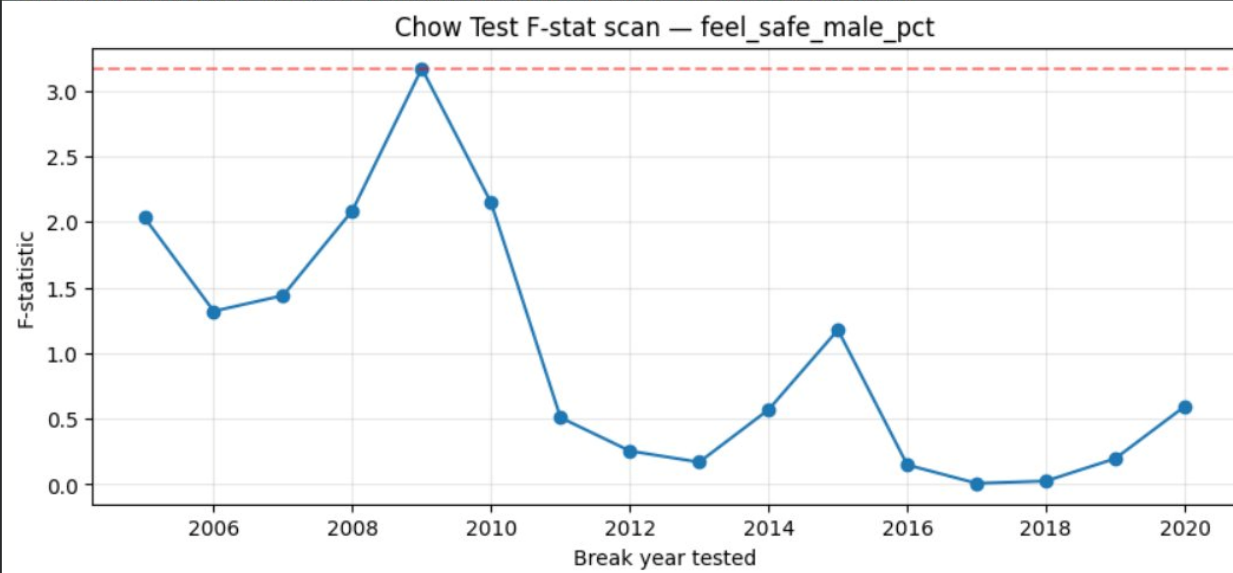
Most likely structural break in [violent_crime_worry_pct]: 2005 ($F=9.7995$, $p=0.0013$)

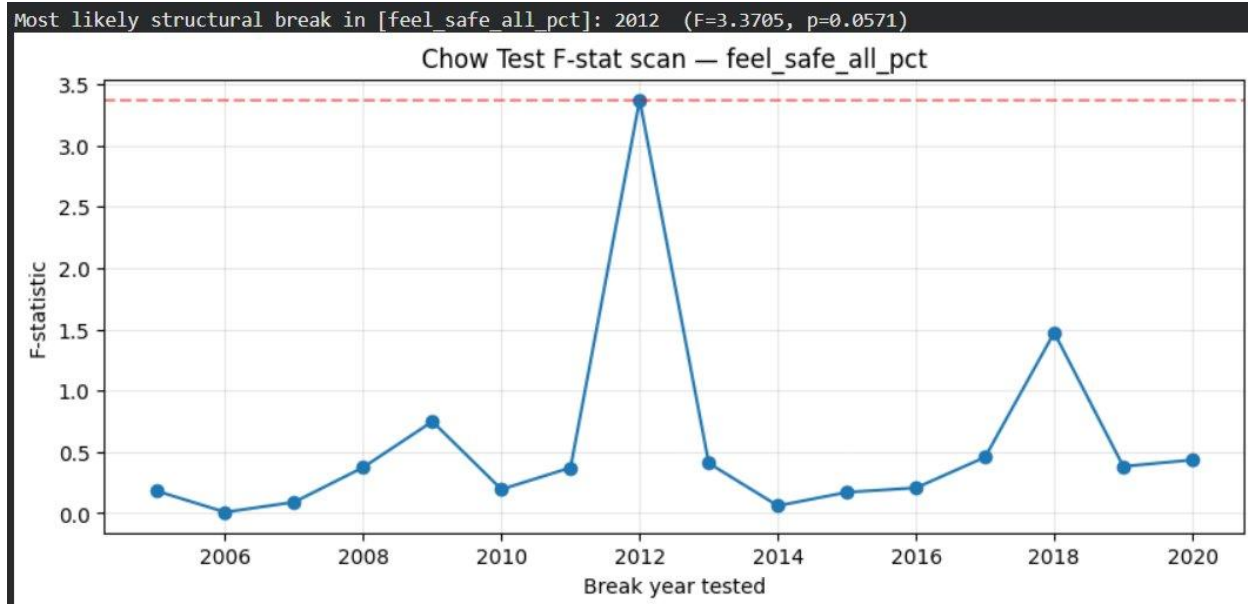


Most likely structural break in [feel_safe_female_pct]: 2012 ($F=6.0814$, $p=0.0096$)



Most likely structural break in [feel_safe_male_pct]: 2009 ($F=3.1711$, $p=0.0661$)





5. Key Findings:

- All worry-about-crime variables show statistically significant structural breaks confirming a **regime shift in public anxiety**.
- Car crime worry: most pronounced break (2012, F=79.5) sharp drop in worry coincides with **falling car crime rates post-2012 recording changes**.
- Burglary and violent crime worry: **breaks at 2011 and 2005 respectively both significant, both correspond to long-term decline acceleration**.
- Feeling safe (female): significant break at 2012 (p=0.01) **reflects meaningful shift in female perception of safety**.
- Feeling safe (male and all): marginal breaks (p~0.06–0.07) **improving trend but less abrupt**.
- Rolling averages confirm: **worry declining consistently 1992–2025; safety perceptions improving, especially for women**.
- **COVID gap (2021–2022) treated as missing data not interpolated, preserving the structural break signal post-pandemic**.